NITIN PANWAR CU24O251575 SEC:B

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os
```

```
# ============================================
# 1. LOAD IMAGE
# ============================================

image_path = "/content/pexels-friededia-30649280.jpg"

image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image not found!")
    print("Check the image path and file name.")
    exit()

print("Image loaded successfully!")
```

```
Image loaded successfully!
```

```
# ============================================
# 3. IMAGE PROPERTIES
# ============================================

height, width, channels = image.shape

print("\n========== IMAGE PROPERTIES ==========")
print("Width       :", width)
print("Height      :", height)
print("Channels    :", channels)
print("Resolution  :", width, "x", height)
print("Data Type   :", image.dtype)
print("Total Pixels:", image.size)
```

```
========== IMAGE PROPERTIES ==========
Width       : 8192
Height      : 5464
Channels    : 3
Resolution  : 8192 x 5464
Data Type   : uint8
Total Pixels: 134283264
```

```
# ============================================
# 4. SAVE IMAGE IN JPEG AND PNG
# ============================================

cv2.imwrite("output_image.jpg", image)
cv2.imwrite("output_image.png", image)

print("\nImages saved successfully:")
print("output_image.jpg")
print("output_image.png")
```

```
Images saved successfully:
output_image.jpg
output_image.png
```

```
# 5. CONVERT TO GRAYSCALE
# ============================================

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

cv2.imwrite("grayscale.jpg", gray)
```

```
True
```

```
# 6. CONVERT TO HSV
# ============================================
```

```
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)

cv2.imwrite("hsv.jpg", hsv)
```

```
True
```

```
# 7. CONVERT TO LAB
# ============================================

lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)

cv2.imwrite("lab.jpg", lab)
```

```
True
```

```
# 8. RESIZE IMAGE
# ============================================

new_width = 400
new_height = 300

resized = cv2.resize(image, (new_width, new_height))

cv2.imwrite("resized.jpg", resized)
```

```
True
```

```
# 9. ROTATE IMAGE
# ============================================

# Rotate 90 degrees clockwise
rotated = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)

cv2.imwrite("rotated.jpg", rotated)
```

```
True
```

```
# 10. HORIZONTAL FLIP
# ============================================

horizontal_flip = cv2.flip(image, 1)

cv2.imwrite("horizontal_flip.jpg", horizontal_flip)
```

```
True
```

```
# 11. VERTICAL FLIP
# ============================================

vertical_flip = cv2.flip(image, 0)

cv2.imwrite("vertical_flip.jpg", vertical_flip)
```

```
True
```

```
# 12. IMAGE COMPLEMENT / NEGATIVE
# ============================================

negative = 255 - image

cv2.imwrite("negative.jpg", negative)
```

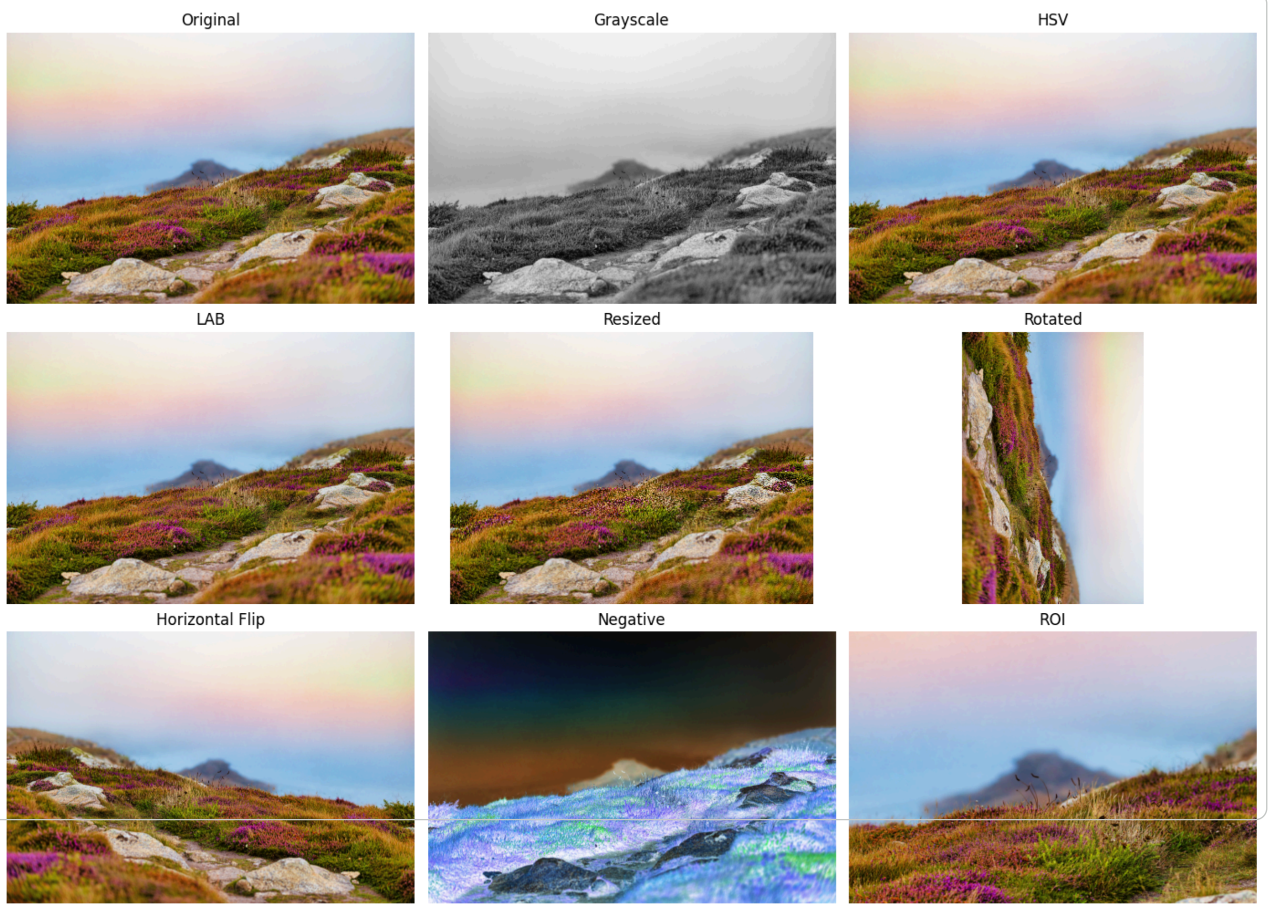
```
True
```

```
# 13. CROP REGION OF INTEREST (ROI)
# ============================================

# Get image dimensions
h, w = image.shape[:2]

# Define ROI
# Format: image[y1:y2, x1:x2]

x1 = int(w * 0.25)
x2 = int(w * 0.75)
```

```
y1 = int(h * 0.25)
y2 = int(h * 0.75)

roi = image[y1:y2, x1:x2]

cv2.imwrite("roi.jpg", roi)
```

```
True
```

```
# 14. DISPLAY ALL PROCESSED IMAGES
# ============================================

plt.figure(figsize=(14, 10))

# Original
plt.subplot(3, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis("off")

# Grayscale
plt.subplot(3, 3, 2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale")
plt.axis("off")

# HSV
plt.subplot(3, 3, 3)
plt.imshow(cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB))
plt.title("HSV")
plt.axis("off")

# LAB
plt.subplot(3, 3, 4)
plt.imshow(cv2.cvtColor(lab, cv2.COLOR_LAB2RGB))
plt.title("LAB")
plt.axis("off")

# Resized
plt.subplot(3, 3, 5)
plt.imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))
plt.title("Resized")
plt.axis("off")

# Rotated
plt.subplot(3, 3, 6)
plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))
plt.title("Rotated")
plt.axis("off")

# Horizontal Flip
plt.subplot(3, 3, 7)
plt.imshow(cv2.cvtColor(horizontal_flip, cv2.COLOR_BGR2RGB))
plt.title("Horizontal Flip")
plt.axis("off")

# Negative
plt.subplot(3, 3, 8)
plt.imshow(cv2.cvtColor(negative, cv2.COLOR_BGR2RGB))
plt.title("Negative")
plt.axis("off")

# ROI
plt.subplot(3, 3, 9)
plt.imshow(cv2.cvtColor(roi, cv2.COLOR_BGR2RGB))
plt.title("ROI")
plt.axis("off")

plt.tight_layout()
plt.show()
```


Original
Grayscale
HSV
LAB
Resized
Rotated
Horizontal Flip
Negative
ROI